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Manual: P.H.A.A.S.T

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1 Usage

PHAAST has one main non-optional and positional arguments, `STOICHIOMETRY`, which defines the atoms that will compose the searched minima, and flags that will adjust the search parameters. The command can be summarized in:

```
1 phaast [OPTIONS] STOICHIOMETRY
```

1.1 Options

All other option arguments can be found on the help section, which is shown with the use of the option `-h`:

```
1 -h, --help                show this help message and exit
2
3 --return-number RETURN-NUMBER
4   Number of least energy molecules to return at the end of the algorithm (
   default = 50)
5
6 -pop, --population-size POPULATION-SIZE
7   Number of molecules in the population (default = 5000)
8
9 -c, --charge CHARGE       Structure's charge (default = 0)
10
11 -t, --threads THREADS
12   Number of threads to be used by the algorithm (default = number of
   threads on cpu)
13
14 -xtbt, --xtb-threads XTB-THREADS
15   Number of threads to be used by each -t xtb instance (default = 4)
16
17 --xtb-path XTB-PATH
18   Path to be used for the xtb binary (default = xtb)
19
20 -et, --energy-threshold ENERGY-THRESHOLD
21   Maximum energy difference [in hartree] so molecules are considered alike
   (default = 0.001)
22
23 -gt, --geometry-threshold GEOMETRY-THRESHOLD
24   Maximum geometry difference so molecules are considered the same so one
   of them is discarded [must be a value between 0 and 1] (default = 0.87)
25
26 --mut-disp-w MUT-DISP-W
27   Mutation displacement weight (default = 3)
28
29 --mut-perm-w MUT-PERM-W
30   Mutation permutation weight (default = 2)
31
32 --mut-twist-w MUT-TWIST-W
33   Mutation twist weight (default = 1)
34
35 --crov-w CROV-W
36   Crossing-over weight (default = 1)
37
38 --migr-w MIGR-W
39   Migrator weight (default = 1)
40
41 --mut-disp-min MUT-DISP-MIN
42   Minimum distance (A) to be used in mutation of atomic displacement (
   default = 1.3)
43
44 --mut-disp-max MUT-DISP-MAX
45   Maximum distance (A) to be used in mutation of atomic displacement (
   default = 2.3)
46
```

```

47 --mut-disp-num MUT-DISP-NUM
48   Number of atoms to move in mutations of atomic displacement (default =
    1)
49
50 --mut-perm-num MUT-PERM-NUM
51   Number of permutations in mutations of that type (default = Half the
    number of atoms)
52
53 --mut-twist-min-angle MUT-TWIST-MIN-ANGLE
54   Minimum angle to be used in twist mutations (default = 90)
55
56 --mut-twist-max-angle MUT-TWIST-MAX-ANGLE
57   Maximum angle to be used in twist mutations (default = 270)
58
59 -loops, --end-loop-number END-LOOP-NUMBER
60   Number of loops the algorithm will run with the least energy molecule
    until a new one is found, terminating it (default = 11)
61
62 --comparison-algorithm COMPARISON-ALGORITHM
63   Algorithm to be used for structures comparison, the recommends usage is:
64   - "grigoryan-springborg" for clusters;
65   - "bonding-length" for organic and general shaped structures (check --
    comparison-bonding-tolerance too);
66   (default = grigoryan-springborg)
67
68 --comparison-bonding-tolerance COMPARISON-BONDING-TOLERANCE
69   Distance tolerable so two atoms can be considered bonded within the
    molecule comparison algorithm and bonding tests
70
71 --remove-unbonded REMOVE-UNBONDED
72   Choose to not remove unbonded structures from genetic algorithm
    population
73   - "always": Remove unbonded in every loop;
74   - "final": Remove only from final population to not contaminate
    results;
75   - "never": Do not remove;
76   (default = always)
77
78 -v, --verbose

```

1.1.1 Return number

Changing the `return-number` will return more or less structures at the end of the algorithm, there's no additional overhead to that, since the structures have already been optimized up to that point. It is useful when the calculator used to optimize the structures is not so accurate on the energy values and a better optimization (on a more refined functional/basisset) is needed at the end of the algorithm, hence justifying a smaller return number.

1.1.2 Population size

Despite how it sounds, this argument doesn't define a cap or a direct control on the population's size, it actually defines how many structures will be generated in each generation, respecting the **Weights** defined at the optional arguments.

1.1.3 Threads

Number of processes (or threads if you are using a GIL-disabled python instance) used for paralelizing the expensive tasks on the genetic algorithm.

1.1.4 Weights

All the `--*-w` arguments define the proportions of how each operation will be applied. For instance, `--mut-disp-w`, `--mut-perm-w` and `--mut-twist-w` define all mutation weights, if they're the same value, all mutations will occur in the same proportion.

1.1.5 Benchmarks

A user can run a benchmark to measure the success rate and speed of the framework usign the `--benchmark-structure minimum.xyz` flag, which will run `--benchmark-num` benchmarks and provide how many of those were able to find the `minimum.xyz` structure among the `--return-number` minima of each run.